

Landuse/Landcover Change Detection Analysis using Geospatial Techniques: A Case Study of Simhadripuram Mandal, Kadapa District, Andhra Pradesh, India

S. Srinivasa Gowd^{1,*}, C. Krupavathi², B. Pradeep Kumar³, P. Ravi Kumar⁴

Abstract

The focus of this research is to distinguish between shifting land-use patterns and their geo-environmental implications in Simhadripuram Mandal, YSR Kadapa district, Andhra Pradesh, India. While the terms Landuse and Landcover (LULC) are sometimes used interchangeably, they have distinct natural meanings. Landuse refers to the function of a piece of land, such as mining, agriculture, or habitation. Landcover refers to the ground's surface cover, which might include vegetation, water, barren soil, and so on. Landuse is a notion that has a lot of qualities. It refers to people's actions on land that are directly related to the land. Landcover, on the other hand, refers to the vegetation that covers the surface of the land. The changes in LULC in Simhadripuram Mandal during the last ten years were studied using satellite imageries. LULC changes have been mapped using remote sensing and GIS modeling. The study is characterized in to five classifications, they are vegetation, fallow land, barren land, waterbodies, and built-up-land. The results reveal from 2009–10 to 2019–20, waterbodies are slightly increased from 2010 to 2020, is 9.29 km². Vegetation is increased positively for the past decade to 25.16 Km². The unirrigated land has been decreased to 27.1 km². Built-up-land is increased from the last decade; it is noted as 22.77 km². Fallow land has been decreased to 30.12 km².

Keywords: Landuse, landcover, supervised classification, remote sensing, GIS

INTRODUCTION

Land use and land cover data are critical inputs for a variety of developmental, environmental, and resource planning applications, as well as regional and global scale process models. This is rather understandable given that the two terms are barely connected and, in some cases, overlap in scope. Landcover simplifies a faultless arrival of the environmental balance between parent rock, climatic ailment, soil, and plant in this natural condition. Landcover is classified into a variety of categories, including vegetation, bare soil, rock outcrops, damp and water bodies, and so on. Landcover is the outcome of observation, to put it simply. Land use also refers to land cover but in terms of its socioeconomic feasibility and international applicability [1–3]. This contrasts sharply with land cover, which, as previously indicated, is the most expressive and deals with physical observations. Land use can vary in nature and intensity, depending on the functions it serves as well as physical observations [4–7]. Because of people's deliberate participation in acquainting the natural land cover to their assistances, land usage differs from land cover.

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Agricultural, industrial, recreational, residential, and other uses are all carried out on land. A region's land use/land cover pattern is the result of

natural and socioeconomic elements, as well as man's usage of them across time and space. Due to massive agricultural activity and population pressure, the land is becoming limited. The natural vegetative cover types that describe a certain area are referred to as land cover [8, 9]. These are usually a reflection of the local climate and landforms, but they can be influenced by human activity as well.

Geo-environment is important not just from an environmental perspective, but also in terms of social economy and human perception. Because of the degradation of the environment and human health in an urban setting, natural and human-induced environmental changes are of concern today. The spatial environments have been underestimated, and we are still far from realizing its maximum potential. An understanding of Earth's resource assets and their undercurrents is essential for resolving problems in a variety of disciplines, including soil, air, and water contamination, soil degradation, waste disposal, and construction provisions and foundations [10]. Other aspects of research, such as the progression and alleviation of natural and man-made threats and jeopardies, land protection and renovation, landscape, and urban planning, ecosystem inventories, and natural heritage assessments, all require an understanding of geological and geomorphological processes [11–14].

Remote sensing has transformed into a valuable technique for developing and comprehending the earth's global physical processes. The usage of satellite data has recently evolved to take advantage of the growing quantity of geographical data available in conjunction with GIS to aid interpretation. GIS is a computer hardware and software system for acquiring, storing, retrieving, manipulating, analyzing, and presenting geographically related (spatial) data to enhance development-oriented management and decision-making processes. "Geographical Information System (GIS) epitomizes a commanding set of tools for gathering, filament, recovering, and presenting spatial data from the real world. Remote Sensing (RS) from airborne and space-borne platforms delivers appreciated data for mapping, geo-environmental monitoring. The incorporation of RS and GIS modeling technologies applied to the field of environmental protection is important and decision making" [15–17].

Study Area

The research area Simhadripuram is fallen in the semi-arid region of YSR Kadapa district, Andhra Pradesh, India. It has a geographical area of 298.98 sq.km and study area lies amongst the Northern latitude of 14° 30' 00" and 14° 45' 00" and the Eastern longitude 78° 05' 00" and 78° 15' 00" (Figure 1). The study part is situated in the Survey of India (SOI) topographical map of 57 J/02. Temperature varies from 27°C to 40°C. The Kadapa receives an average annual rainfall of 759 mm. The YSR district is appropriately named the Pennar Area since the Pennar River and its tributaries drain virtually the whole district. Under the effect of significant rainfall from cyclonic storms in the Bay of Bengal, the district's rivers and streams are primarily ephemeral.

MATERIALS AND METHODS

Data Acquired

1. SOI (Survey of India) topo sheet no. 57 J/02
2. Landsat imageries of three dissimilar periods i.e., 2009–2010 and 2019–2020 have been taken free from the USGS earth explorer site [18] and NRSC site [19].
3. Software used
 1. ArcGIS 10.3
 2. ERDAS imagine 2014

Processing of Data and Preparation of LULC Map

Data corrections have to be made for image preprocessing, they are likely atmospheric, image matching and mosaic and geometric etc., with the aid of topographical maps reference using ArcGIS software. After that for the generation of the LULC map, Supervised classification was conceded out

in ERDAS imagine software [20] (Figure 2). Visual interpretation like tone, the texture is cross-checked with the assistance of the Google Earth engine in this software with carrying out signature editor option is selected in raster tool. Before going to the signature editing, satellite imageries of the study area of the three different years were taken i.e., 2009–10 and 2019–20 and layer stacking is done and then proceed to subset option for selecting the study area boundary and these imageries were selected and done as new signature files. Every feature or class has been divided and done by visual observation of the satellite imageries. Figure 3 can depict the tone and textures of the feature. In this supervised classification, a specific signature file has to be prepared for each class. According to signature classification, five categories (six features) have been identified in the study area, namely (i) Vegetation (ii) Fallow land (iii) Unirrigated lands (iv) Unclassified (which is not shown in the LULC maps because in supervised classification it should be considered as zero) (v) Water bodies (vi) Built-up-land. The following flowchart demonstrates the adopted methodology for the LULC map preparation (Figure 2).

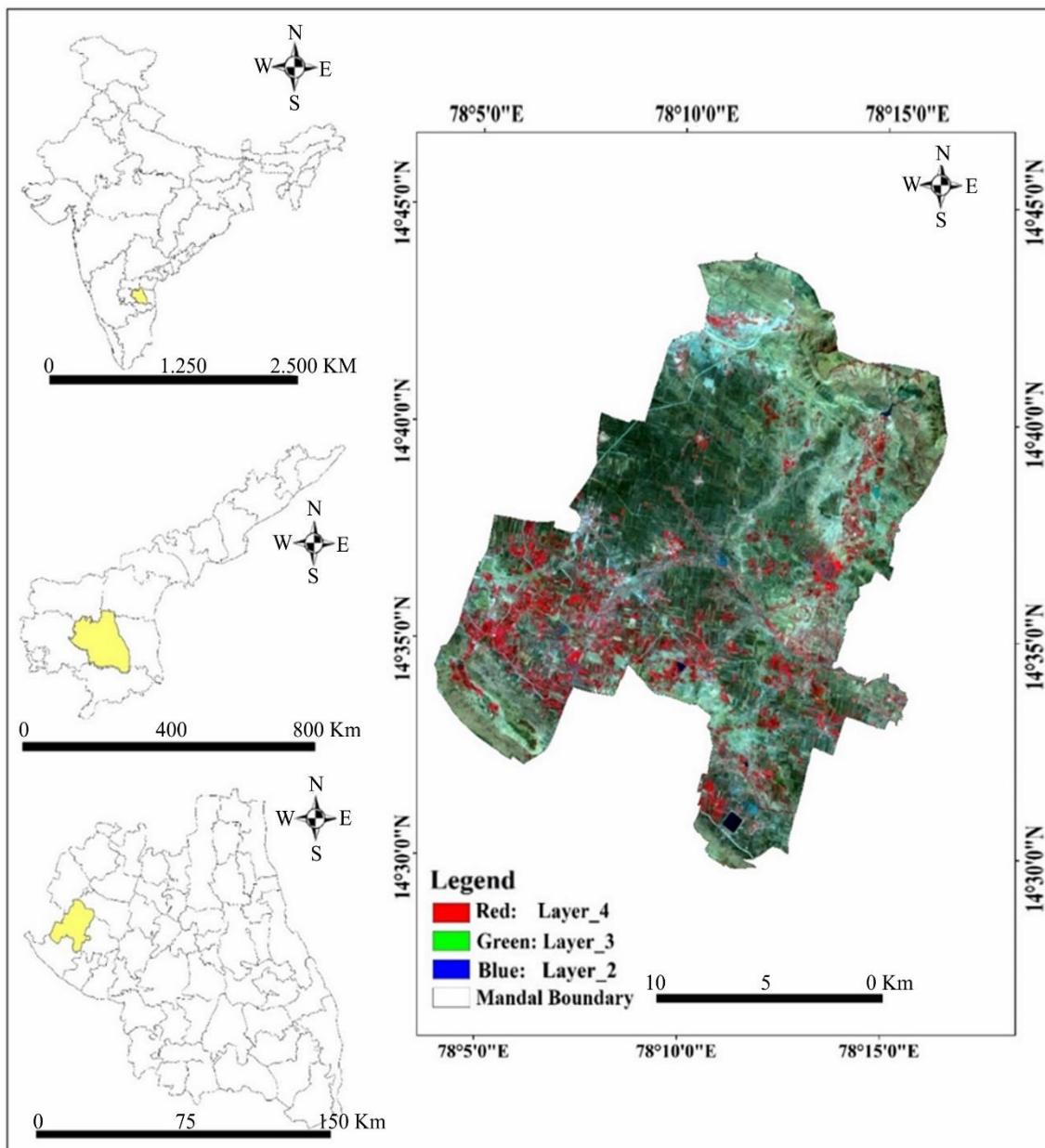


Figure 1. Location map of the study area.

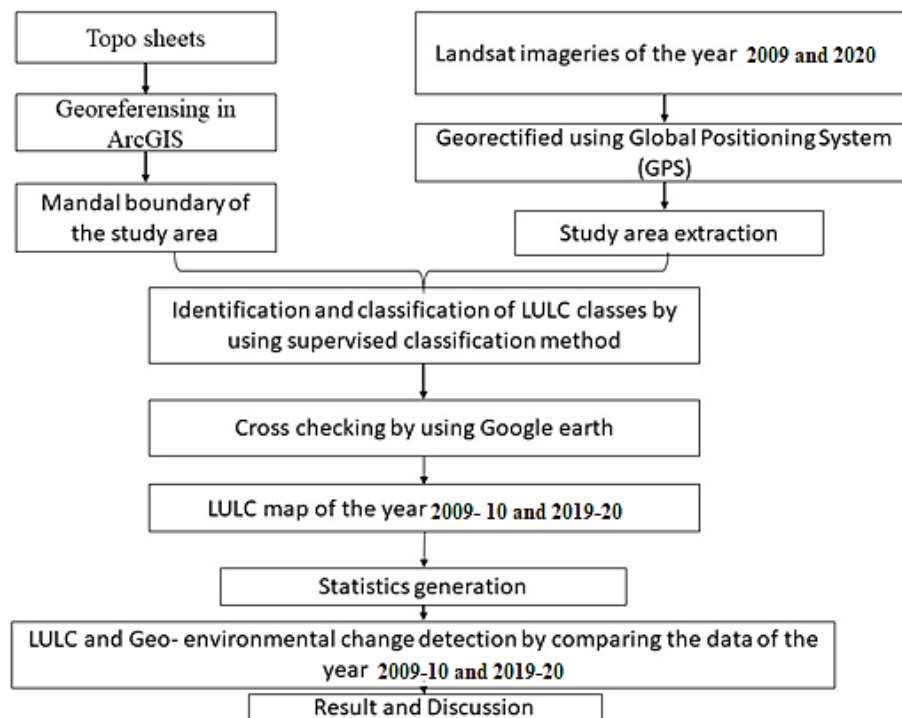


Figure 2. Flowchart for the methodology.

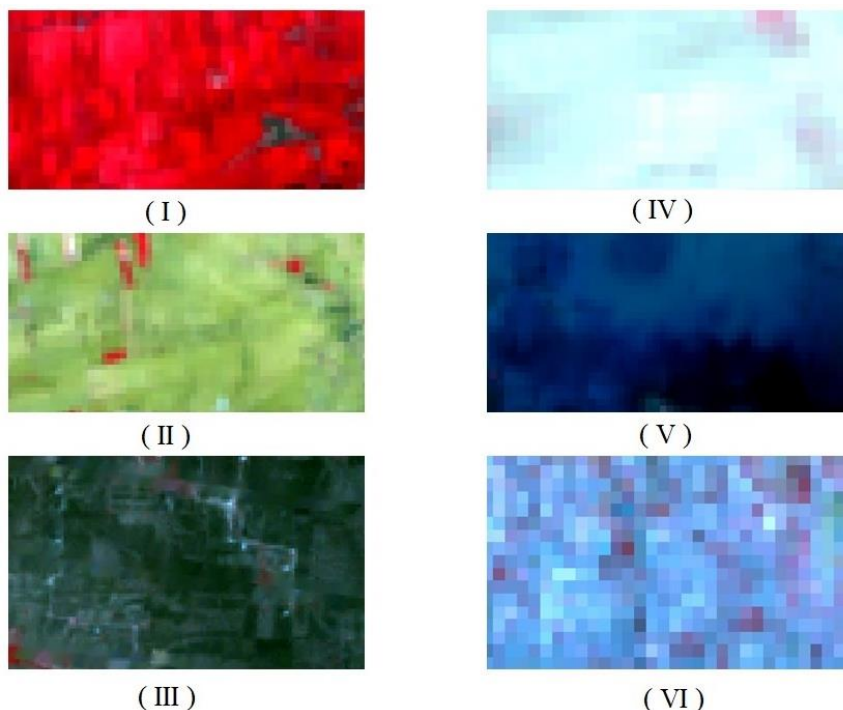


Figure 3. Tone/Texture of selected features in Landsat imageries, (i) Vegetation (ii) Fallow land(iii) Unirrigated lands (iv) Unclassified (v) Water bodies (vi) Built-up-land.

RESULTS AND DISCUSSION

Geology of the Study Area

The study area geologically belongs to Kadapa formations, those are the oldest rocks of the area belonging to the Late Achaean or Early Proterozoic era which is succeeded by rocks of the Dharwarian Age, and both are traversed by dolerite dykes. The older rocks are overlain by rocks of

the Cuddapah Supergroup and Kurnool Group belonging to the Middle and Upper Proterozoic Age. The main lithological units consist chiefly of Tadipatri Formation of Dolomite, Quartzite, Volcanic Flows, and Shale. Figure 4 depicts the geology of the study area.

Geological Succession		
GROUP	FORMATION	LITHOLOGY
Kurnool Group	-----Unconformity-----	
Chitravati Group	Tadipatri Formation	Dolomite, Quartzite, Volcanic Flows and Shale.
	-----Unconformity-----	
Peninsular Gneissic Complex	Granite Gneisses, Schist, Granitoids with acidic and basic intrusive	

Geomorphology of the Study Area

Geomorphology is the study of landforms, the restoration of the processes that caused them, and the investigation of the effect of tectonics in time and space. The inventory and categorization of landforms are part of the geomorphological mapping process. Each landform is influenced by the environment, which includes soil cover, hydrology, and hydrogeology, as well as its composition and depth of weathering structural frame. The landforms are divided into three categories depending on their origin, relief slope factor, and surface cover. Figure 5 depicts the landforms that occur in the region, which are classified as Pediplain, residual hill, and structural hill.

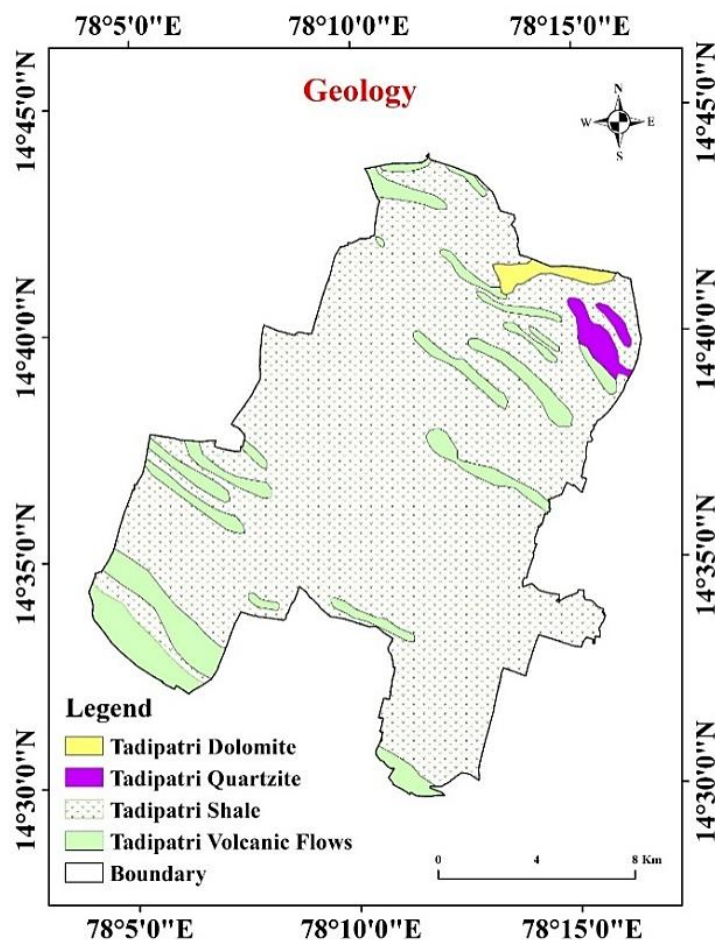


Figure 4. Geological map of the study area.

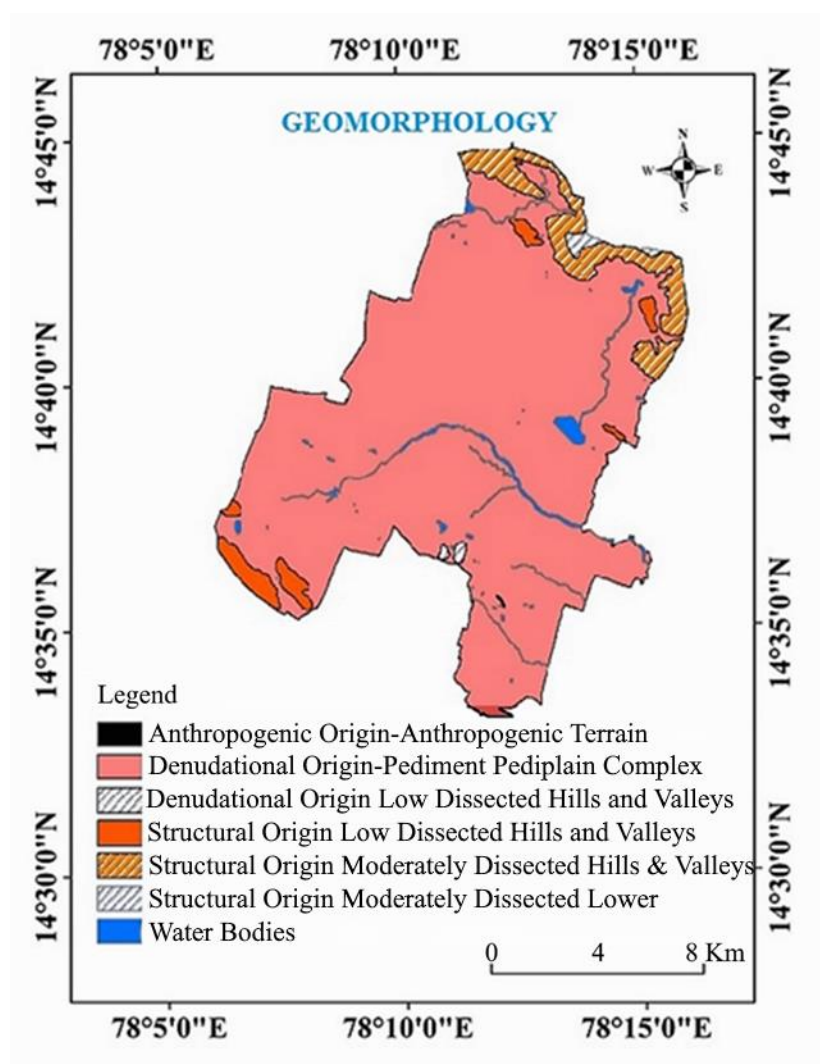


Figure 5. Geomorphology map of the study area.

Landuse/Landcover Analysis (LULC)

The data gained through the analysis of Landsat imagery are placed in Table 1 and Table 2, the demonstrated resultant Figures 6 and 7. These figures provide information about the land use/cover pattern of the target area. Figure 6 represents the status of the LULC map of the study area during 2009–10. Figure 7 represents the LULC during 2019–20 of the study part. Bar diagram showing the percentage of change in LULC and geo-environmental changes among the year 2009–10 and 2019–20 (Figure 8). Figure 9 shows the graphical representation of LULC and its impact on geo-environment changes in the percentage during the decade i.e., 2009–10 and 2019–20.

Table 1. 2009–10 and 2019–20 resultant changes in LULC

LULC categories	2009–10		2019–20	
	Km ²	Percentage	Km ²	Percentage
Waterbodies	1.36	0.45	10.65	3.6
Vegetation	53.07	17.8	78.23	26.22
Unirrigated lands	121.41	40.7	94.31	31.61
Built up land	32.13	10.77	54.9	18.4
Fallow land	90.33	30.28	60.21	20.17
	298.3	100	298.3	100

Table 2. Shows the resultant changes from 2010 to 2020 in LULC changes

LULC categories	Changes from 2010–2020	
	Km ²	Percentage
Waterbodies	9.29	3.15
Vegetation	25.16	8.42
Unirrigated lands	-27.1	-9.09
Built up land	22.77	7.63
Fallow land	-30.12	10.11

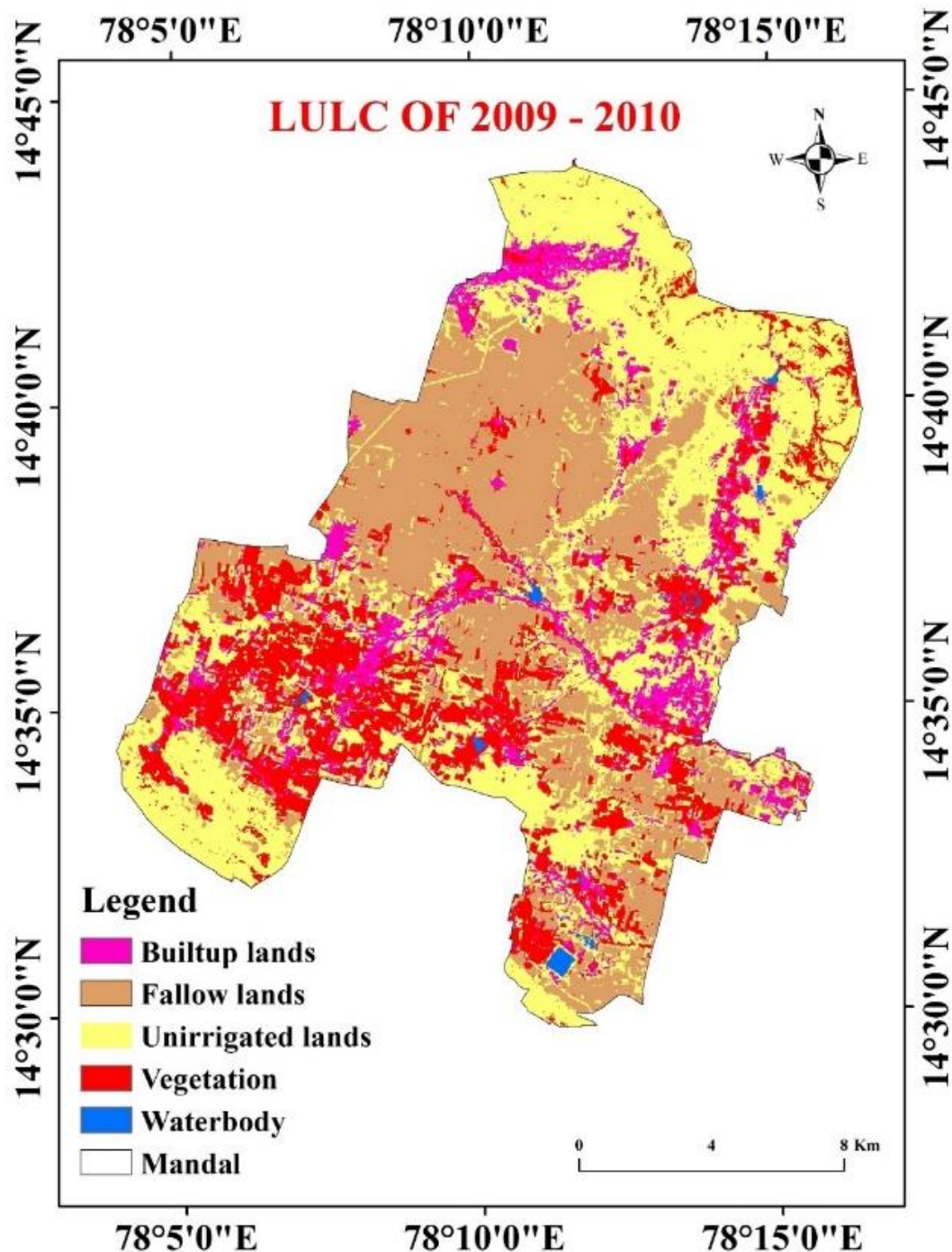


Figure 6. LULC changes in the year 2000.

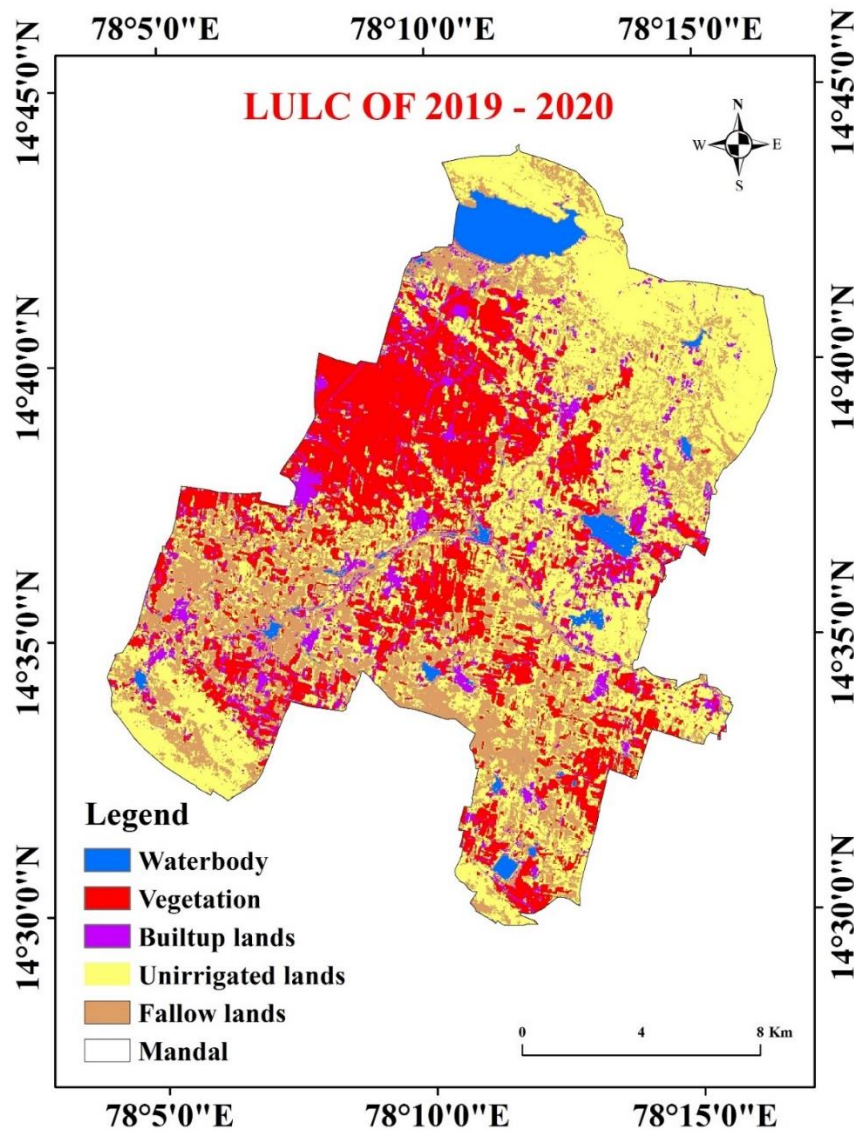


Figure 7. LULC changes in the year 2010.

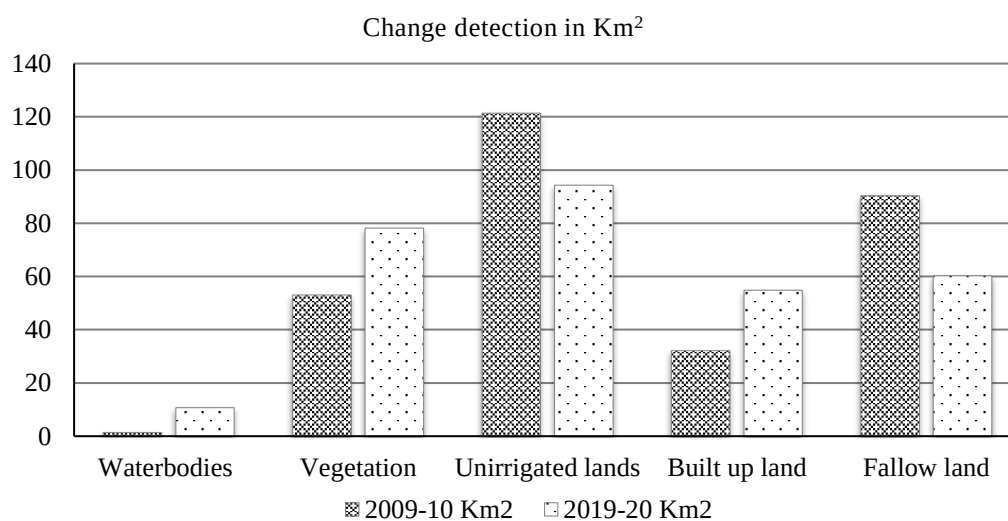


Figure 8. Graphical Representation of LULC changes from 2009–10 and 2019–20.

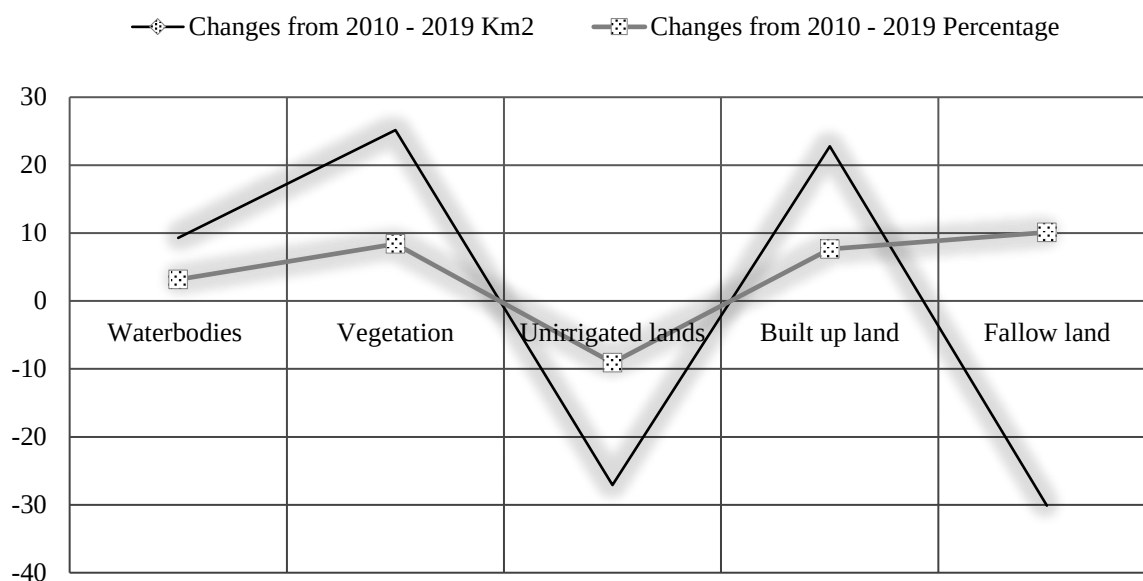


Figure 9. Resultant changes in km² and in percentage.

Waterbodies

Any considerable collection of water, usually on a planet's surface, is referred to as a body of water or waterbody (commonly written water body). Some bodies of water, such as rivers and streams, gather and transport water, while others, such as lakes and oceans, largely keep it. Including all water bodies (rivers, lakes, gravels, streams, canals, and reservoirs). In the year 2009–10 waterbodies are noticed as 1.36 km² (0.45%), and in the year 2019–20, it is noticed as to 10.65 km² (3.6%). Results reveal that the waterbodies are slightly increased from 2010 to 2020, is 9.29 km² (3.15%).

Vegetation

Forest, grassland, tundra, wasteland, and glaciers are the five basic types of vegetation zones. Climate, soil, and the ability of soil to store water, as well as the slope, or angle, of the ground, all influence the sorts of plants that may thrive in a given area [21] (Khan and Jhariya, 2016). The percentage of soil covered by green plants is known as vegetation cover. Plant cycle, crop harvesting, animal grazing, plant trimming, and other factors cause spatial and temporal variations in vegetation cover within the same year. Including all agricultural and forest lands. Usually, vegetation/agricultural texture is red color in the satellite imageries. For this LULC mapping two dissimilar Landsat imageries were collected and done mapping. In the year 2009–10 vegetation 53.07 km² (17.8%) is noticed. When coming to the year 2019–20 it is noted as 78.23 km² (26.22%) is Increased. Finally, results reveal that Vegetation is increased positively for the past decade to 25.16 Km² or 8.42%.

Unirrigated Land

Unirrigated land is defined as, not being supplied with water by irrigation: not irrigated unirrigated land/crops. In the year 2009–10 Unirrigated land is 121.41 km² (40.70%) and in the year 2019–20, it is decreased to 94.31 km² (31.61%). Finally, results show that the unirrigated land has been decreased to 27.1 km² (9.09%). Because of increasing the availability of water, most of the people cultivated their lands hence vegetation is increased in the study area, this leads to a decrease in the Unirrigated lands.

Built-up-land

An urban area/Built-up land is the region surrounding a city. Most inhabitants of urban areas have nonagricultural jobs. Urban areas are very developed, meaning human structures are dense such as houses, commercial buildings, roads, bridges, and railways. "Urban area/Built-up land" can refer to towns, cities, and suburbs.

Including all residential, commercial, industrial, and transportation. Now a day's increasing the built-up land/urbanization is very common, because of the increase in population. In the year 2009–10, it is noticed as 32.13 km² (10.77%) and it is increased in the year 2019–20 to 54.9 km² (18.40%). Results reveal that built-up-land is increased from the last decade; it is noted as 22.77 km² (7.63%).

Fallow Land

A piece of land that is normally used for farming but that is left with no crops on it for a season to let it recover its fertility is an example of land that would be described as fallow. True fallow includes bare fallow (tillage begins in the summer or autumn after harvesting the previous crop), early fallow (the soil is plowed in the spring of the next year), and strip fallow (with strips of corn, sunflower, mustard, and other tall-stalked plants planted in the field to retain snow). Including all land areas that exposed soil and the barren area included by a human. Fallow land was estimated to be 90.33 km² (30.28%) in 2009–10, but it has declined to 60.21 km² in 2019–20. (20.17%). Finally, results show that the fallow land has been decreased to 30.12 km² (10.11%) between the years 2010–2020.

CONCLUSION

To use an integrated technique of remote sensing and GIS modeling, this study evaluated land use patterns, significant improvement movements, and environmental implications of land-use change in the Simhadripuram region over the last ten years. LULC changes were significant from 2009–10 to 2019–20, according to the findings. There is a positive sign that the increase in waterbodies is noticed (1.12%). On the other hand, vegetation, fallow land, and barren land are decreased (0.55%, 0.76%, and 3.39%). Built-up-land is also decreased (1.39%). Most of the sand migration is noticed in this north-western part. For a living, the majority of the residents in these communities must migrate to neighboring villages or towns. As a result, the environment's ecology becomes unbalanced. The RS and GIS make suggestions on techniques for monitoring, approximating, evaluating, and supervising environmental threats in order to preserve lives and society.

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